The Augmented Dold-Kan Correspondence

Science Research Project (SCIE30001) Report

Lochlan (Tresize) Eastwood 1453164*

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Abstract

The Dold-Kan correspondence establishes an equivalence between the category of simplicial abelian groups and chain complexes over abelian groups. That is, $\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab}) \simeq \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$. In this article we consider the augmented case, $\mathbb{\Delta}_\eta = \mathbb{\Delta} \sqcup [-1]$ and use Dold-Kan to demonstrate the equivalence $\mathrm{Fun}({\mathbb{\Delta}_\eta}^{\mathrm{op}}, \mathbf{Ab}) \simeq \mathbf{Ch}_{\geq -1}(\mathbf{Ab})$. We call this equivalence the *augmented Dold-Kan correspondence*.

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*Available at https://gitlab.unimelb.edu.au/leastwood/Science-Research

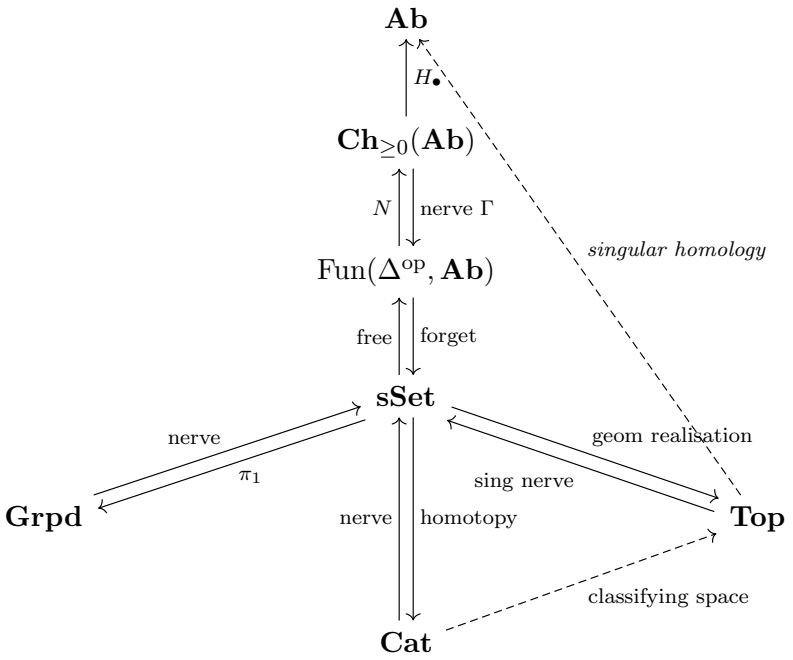
Email: leastwood@student.unimelb.edu.au
Supervisor: Nora Ganter
Co-assessor: Lior Yanovski
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1 Introduction

> [Simplicial sets] are indispensable; and at times discussions are greatly clarified when described in this language.
>
> Benson, *Representations and Cohomology: Volume 2, Cohomology of Groups and Modules* 1991.

In this article we will study the category of *simplicial abelian groups*, $\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab})$, a subcategory of the category of simplicial sets, **sSet**. Simplicial sets are, in some sense, unifying in category theory, topology and homotopy theory. One sense in which this is true is the following diagram.[1]



Each pair of arrows is an adjunction. In fact, $\mathbf{Top} \leftrightarrows \mathbf{sSet}$ and $\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab}) \leftrightarrows \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$ extend to Quillen equivalences [Hug22]. The adjunction $\mathrm{Fun}(\mathbb{\Delta}, \mathbf{Ab}) \leftrightarrows \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$ is the Dold-Kan correspondence.

The content of this article sits in the portion of the diagram above **sSet**, namely around the Dold-Kan correspondence. The Dold-Kan correspondence is central to the many stories encompassed in the above diagram, in particular, in homotopy theory, since it extends to

[1] Diagram adapted from David Wedgmann after a discussion with Adam Monteleone

a Quillen equivalence. However, the correspondence is built from the standard simplex $\mathbb{\Delta}$, which lacks *monoidal structure*. This is rectified by *augmenting*, which involves adding an initial object to $\mathbb{\Delta}$ to produce the monoidal category $\mathbb{\Delta}_\eta$.

One can ask how much of the above diagram is recovered in this augmented world. The main theorem of this article is that the Dold-Kan correspondence has an augmented analogue, which we call the *augmented Dold-Kan correspondence* (Theorem 2.7).

Theorem 1.1 (Augmented Dold-Kan correspondence)**.** *There is an equivalence of categories*

$$\mathrm{Fun}({\mathbb{\Delta}_\eta}^{\mathrm{op}}, \mathbf{Ab}) \simeq \mathbf{Ch}_{\geq -1}(\mathbf{Ab}).$$

The punch line of this article (besides this theorem) is that *comma categories* are the lingua franca in this augmented world. Namely, they allow us to split up $\mathrm{Fun}({\mathbb{\Delta}_\eta}^{\mathrm{op}}, \mathbf{Ab})$ into $\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab})$ plus some additional structure. We will then use the classical Dold-Kan correspondence to pass down into $\mathbf{Ch}_{\geq 0}(\mathbf{Ab})$ plus some additional structure, which will collect nicely into $\mathbf{Ch}_{\geq -1}(\mathbf{Ab})$.

1.1 Assumed knowledge and recommended literature

We assume the reader is familiar with basic category theory, for example concepts developed in a course in algebraic topology: categories, functors, natural transformations and chain complexes. Those unfamiliar with comma categories should consult Appendix A.3.

Literature is recommended throughout. For general recommendations, [Mac13] is a good introduction to category theory, as is [Rie16]. [GJ12] is the recommendation for simplicial sets; their section on the Dold-Kan correspondence and applications to homotopy theory are essential.

2 The Augmented Dold-Kan Correspondence

We will define simplicial abelian groups, construct the normalised chain complex functor N and build up to the classical Dold-Kan correspondence (Theorem 2.6). We will then pivot to augmented simplicial abelian groups, construct an analogue to N, called N^a, state the augmented Dold-Kan correspondence (Theorem 2.7) and give its proof (Subsection 2.4).

Let $\mathbb{\Delta}$ be the category of finite, totally ordered sets, together with order-preserving maps. Write objects in $\mathbb{\Delta}$ as $[n] = \{0 \dots n\}$ with the usual order. Sometimes this order is represented by arrows

$$[n] = 0 \to 1 \to \cdots \to n.$$

Write $\mathbb{\Delta}_\eta$ to be $\mathbb{\Delta}$ with an initial object $[-1]$ added.[2] We call $\mathbb{\Delta}$ the simplex category and $\mathbb{\Delta}_\eta$ the *augmented* simplex category.

Definition 2.1. A simplicial object of a category $\mathcal{C}$ is a functor

$$X_\bullet : \mathbb{\Delta}^{\mathrm{op}} \longrightarrow \mathcal{C}. \tag{2.1}$$

An *augmented* simplicial object is, similarly, a functor ${\mathbb{\Delta}_\eta}^{\mathrm{op}} \to \mathcal{C}$. Write $X_n \in \mathcal{C}$ for the image of $[n]$ under either functor. It will be useful to identify $X_\bullet$ with $(X_n)_{n\geq 0}$.

In $\mathbb{\Delta}$ and $\mathbb{\Delta}_\eta$ there are maps

$$d^i_n : [n-1] \longrightarrow [n], \quad (0 \to \cdots \to n-1) \mapsto (0 \to \cdots \to i-1 \to i+1 \to \cdots \to n)$$

and

$$s^j_n : [n+1] \longrightarrow [n], \quad (0 \to \cdots \to n+1) \mapsto (0 \to \cdots \to j \to j \to \cdots \to n)$$

$0 \leq i, j \leq n$ (i.e. composing $i-1 \to i \to i+1$ and inserting the identity $j \to j$ [GJ12]). These are called the coface and codegeneracy maps, respectively. When we pass to $\mathcal{C}$ via the functor ${\mathbb{\Delta}_\eta}^{\mathrm{op}} \to \mathcal{C}$, we call these maps d^n_i, s^n_j, the face and degeneracy maps, respectively.

We can represent the data of a simplicial object ${\mathbb{\Delta}_\eta}^{\mathrm{op}} \to \mathcal{C}$ as the following diagram.

$$X_{-1} \dashleftarrow X_0 \;\substack{\longleftarrow \\ \longrightarrow \\ \longleftarrow}\; X_1 \;\substack{\longleftarrow \\ \longleftarrow \\ \longrightarrow \\ \longleftarrow \\ \longrightarrow \\ \longleftarrow}\; X_2 \qquad \dots$$

The dotted arrow represents the arrow $[-1] \to [0]$ in $\mathbb{\Delta}_\eta$, which is not present in $\mathbb{\Delta}$. The dotted arrow $X_0 \to X_{-1}$ is denoted ϵ and is the coequaliser of the face maps

$$\epsilon d^1_0 = \epsilon d^1_1$$

(see [Wei94, Chapter 8.4]).

There are many *simplicial identities* (see [GJ12]); we make use of only one of them.

[2] It's more common to see $\mathbb{\Delta}_+$ written here, but this is ambiguous, see here.

Lemma 2.1 (A cosimplicial identity)**.**

$$d^j_{n+1}d^i_n = d^i_{n+1}d^{j-1}_n \quad \textit{for} \quad i < j.$$

Proof. For $d^j_{n+1}d^i_n$,

$$\begin{aligned}(0 \to \cdots \to n-1) &\mapsto (0 \to \cdots \to i-1 \to i+1 \to \cdots \to n) \\ &\mapsto (0 \to \cdots \to i-1 \to i+1 \to \cdots \to j-2 \to j \to \cdots \to n+1)\end{aligned}$$

where $j - 2 \geq i + 1$. One can show we get the same result for $d^i_{n+1}d^{j-1}_n$. □

Passing through the simplicial object ${\Delta_\eta}^{\mathrm{op}} \to \mathcal{C}$ we obtain the simplicial identity

Corollary 2.1.1.

$$d^n_i d^{n+1}_j = d^n_{j-1}d^{n+1}_i \quad \textit{for} \quad i < j.$$

2.1 Universal monoid

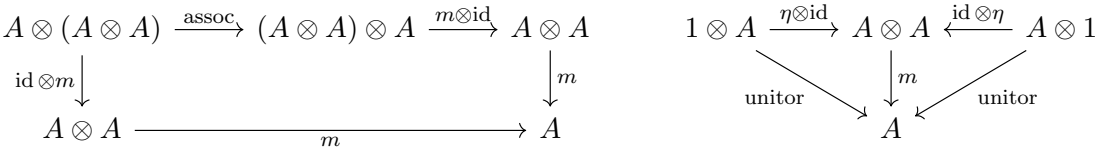
There are a number of reasons for studying Δ_η. As we pointed out, the monoidal structure in Δ_η is one such reason. We will now see why.

One can specify a monoidal structure on Δ_η, for instance by setting $[n] \sqcup [m] = [n+m+1]$ on objects and specifying how $\sqcup$ acts on arrows (see [Mac13, Chapter 7.5]). Note that, although $\sqcup$ is used here, $\sqcup$ is *not* a coproduct.

Definition 2.2 ([Mac13])**.** Let $(M, \otimes, 1)$ be a monoidal category. A *monoid* object in M is an object A with two arrows

$$m : A \otimes A \to A, \quad \eta : 1 \to A \tag{2.2}$$

such that the following commute



(the unitor and associator maps are those that come with $(M, \otimes, 1)$).

Example 2.1. In Δ_η we have the arrows

$$[1] = [0] \sqcup [0] \longrightarrow [0], \quad [-1] \longrightarrow [0]. \tag{2.3}$$

It is easy to check that these arrows satisfy the coherence conditions in Definition 2.2. Hence, [0] is a monoid in Δ_η.

The following theorem allows us to call $\mathbb{\Delta}_\eta$ the universal monoidal category with a monoid object.

Theorem 2.2 ([Mac13])**.** *Let (A, m, η) be a monoid in a strict monoidal category $(M, \otimes, 1)$. There is then a unique arrow with diagram*

$$\begin{array}{ccccccc} \mathbb{\Delta}_\eta & & [1] = [0] \sqcup [0] & \longrightarrow & [0] & \longleftarrow & [-1] \\ {\scriptstyle \exists! F}\,\vdots\downarrow & & \vdots\downarrow & & \vdots\downarrow & & \vdots\downarrow \\ M & & A \otimes A & \longrightarrow & A & \longleftarrow & 1 \end{array}$$

MacLane's coherence theorem then gives that there are such arrows for every monoidal category $(M, \otimes, 1)$.

2.2 The Dold-Kan correspondence

We will construct the *normalised chain complex functor* and discuss its role in the Dold-Kan correspondence.

Definition 2.3. A simplicial abelian group $A_\bullet$ is a simplicial object in the category of abelian groups. That is, a functor

$$\mathbb{\Delta}^{\mathrm{op}} \longrightarrow \mathbf{Ab}.$$

The face maps $d_i^n : A_n \to A_{n-1}$ for $n \geq 0$ and $0 \leq i \leq n$ give us the following lemma.

Lemma 2.3. *There is a chain complex (which we also denote $A_\bullet$)*

$$A_\bullet = (A_n, \partial_{n+1} : A_{n+1} \longrightarrow A_n), \quad \partial_n = \sum_{i=0}^{n} (-1)^i d_i^n.$$

called the Moore complex.

Proof.

$$\begin{aligned}
\partial_{n-1}\partial_n &= \sum_{i=0}^{n-1}(-1)^i d_i^{n-1}\left(\sum_{j=0}^{n}(-1)^j d_j^n\right) \\
&= \sum_{i,j}(-1)^{ij} d_i^{n-1} d_j^n \\
&= \sum_{i<j}(-1)^{ij} d_i^{n-1} d_j^n + \sum_{i\geq j}(-1)^{ij} d_i^{n-1} d_j^n \\
&= \sum_{i<j}(-1)^{ij} d_{j-1}^{n-1} d_i^n + \sum_{i\geq j}(-1)^{ij} d_i^{n-1} d_j^n \quad \text{(Lem 2.1.1)} \\
&= -\sum_{i\leq j}(-1)^{ij} d_j^{n-1} d_i^n + \sum_{i\geq j}(-1)^{ij} d_i^{n-1} d_j^n \quad \text{(reindexing)} \\
&= -\sum_{j\leq i}(-1)^{ij} d_i^{n-1} d_j^n + \sum_{i\geq j}(-1)^{ij} d_i^{n-1} d_j^n \quad \text{(swap } i,j \text{ in first sum)} \\
&= 0.
\end{aligned}$$

□

There is another chain complex one can construct from $A_\bullet$ called the *normalised complex* $NA_\bullet$ (see [Mat11]). Write d_n to mean d_n^n. Let NA_n be the subgroup of A_n killed by each face map d_i^n, $i < n$. Explicitly,

$$NA_n = \bigcap_{i<n} \ker(d_i^n : A_n \longrightarrow A_{n-1}) \subseteq A_n \tag{2.4}$$

with $NA_0 = A_0$. This is a subgroup since the intersection is finite. The boundary maps are

$$(-1)^n d_n : NA_n \longrightarrow NA_{n-1}.$$

Lemma 2.4. *$NA_\bullet = (A_n, (-1)^{n+1} d_{n+1})$ is a chain complex.*

Proof. We are left to check composition of consecutive boundary maps is zero. Indeed,

$$\begin{aligned}
(-1)^n d_n \circ (-1)^{n+1} d_{n+1} &= -d_n d_{n+1} \\
&= -d_n d_n^{n+1} \\
&= 0.
\end{aligned}$$

The second equality is again due to Lemma 2.1.1. The final equality follows by definition of NA_{n+1}. Namely,

$$-d_n d_n^{n+1} : NA_{n+1} \longrightarrow NA_{n-1}$$

and d_n^{n+1} is zero on NA_{n+1}. □

We now define how N acts on arrows. Let $\eta = (\eta_n)_{n\geq 0} : A_\bullet \longrightarrow B_\bullet$ be an arrow in $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab})$. That is, η is a natural transformation. Let η have components η_n for each $n \geq 0$. For any $m, n \geq 0$ and any arrow $[n] \longrightarrow [m]$ we have the diagram

$$\begin{array}{ccc} [n] & & A_n \xrightarrow{\eta_n} B_n \\ \downarrow & & \uparrow \quad\quad \uparrow \\ [m] & & A_m \xrightarrow[\eta_m]{} B_m \end{array} \tag{2.5}$$

Let $N\eta : NA_\bullet \longrightarrow NB_\bullet$ be the chain map with the (restricted) arrows

$$\eta_n|_{NA_n} : NA_n \longrightarrow NB_n.$$

That these restrictions are well defined follows from investigating the natural transformation η on the arrows $d_n^i : [n-1] \longrightarrow [n]$.

We need to verify that squares in

$$\begin{array}{ccccccccc} \ldots & \longleftarrow & NA_{n-1} & \xleftarrow{(-1)^n d_n^A} & NA_n & \xleftarrow{(-1)^{n+1} d_{n+1}^A} & NA_{n+1} & \longleftarrow & \ldots \\ & & \downarrow{\scriptstyle \eta_{n-1}} & & \downarrow{\scriptstyle \eta_n} & & \downarrow{\scriptstyle \eta_{n+1}} & & \\ \ldots & \longleftarrow & NB_{n-1} & \xleftarrow[(-1)^n d_n^B]{} & NB_n & \xleftarrow[(-1)^{n+1} d_{n+1}^B]{} & NB_{n+1} & \longleftarrow & \ldots \end{array}$$

commute. But this follows by 2.5, since face maps $d_n : A_n \longrightarrow A_{n-1}$ come from arrows $[n-1] \longrightarrow [n]$ in Δ. One can show that N is a functor.

There are two important theorems regarding the Moore and normalised complexes. For the first we follow [GJ12, Chapter 3]. Note there is a chain map

$$NA_\bullet \longrightarrow A_\bullet$$

given by the inclusions $NA_n \hookrightarrow A_n$. Namely, the squares in

$$\begin{array}{ccccccccc} \ldots & \longleftarrow & NA_{n-1} & \xleftarrow{(-1)^n d_n} & NA_n & \xleftarrow{(-1)^{n+1} d_{n+1}} & NA_{n+1} & \longleftarrow & \ldots \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \ldots & \longleftarrow & A_{n-1} & \xleftarrow[\partial_n]{} & A_n & \xleftarrow[\partial_{n+1}]{} & A_{n+1} & \longleftarrow & \ldots \end{array}$$

commute since all but 1 of the terms in ∂_n is killed on NA_n.

Theorem 2.5 ([GJ12])**.** *Let $A_\bullet$ be a simplicial abelian group. The chain map $NA_\bullet \longrightarrow A_\bullet$ is a chain homotopy equivalence. In particular,*

$$H_n(NA) \cong H_n(A) \quad (n \geq 0).$$

This is sometimes called the *Eilenberg-MacLane normalisation theorem*.

Theorem 2.6 (Dold-Kan correspondence)**.** *The functor $N : A_\bullet \mapsto NA_\bullet$ is an equivalence between the category of simplicial abelian groups and chain complexes of abelian groups.*

Proof. See [GJ12]. □

This was proven separately by Dold and Kan in 1958 ([Dol58; Kan58]). The result was extended to all abelian categories by Dold and Puppe in 1961 [DP61].

2.3 The augmented Dold-Kan correspondence

We will construct an analogue to the normalised chain complex functor and state the augmented Dold-Kan correspondence (Theorem 2.7).

Recall for a simplicial abelian group $A_\bullet : {\Delta_\eta}^{\mathrm{op}} \longrightarrow \mathbf{Ab}$ we have

$$X_{-1} \xleftarrow{\epsilon} X_0 \;\substack{\longleftarrow\\ \longleftarrow}\; X_1 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow}\; X_2 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow}\; \dots$$

where ϵ coequalises the two face maps

$$\epsilon d_0 = \epsilon d_1.$$

Define the normalised augmented chain complex functor N^a as follows. Let $N^a A_n = NA_n$ for $n \geq 0$ and

$$N^a A_{-1} = A_{-1}$$

It is easy to check

$$A_{-1} \xleftarrow{\epsilon} A_0 \xleftarrow{(-1)d_1} NA_1 \longleftarrow \dots \longleftarrow NA_{n-1} \xleftarrow{(-1)^n d_n} NA_n \longleftarrow \dots$$

forms a chain complex in degrees ≥ -1. In particular, $d_0^1 = 0$ on NA_1 and, hence,

$$\epsilon \circ (-d_1) = \epsilon \circ (d_0^1 - d_1) = 0.$$

Now let $\eta : A_\bullet \longrightarrow B_\bullet$ be an arrow in $\mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathbf{Ab})$. Analogous to the construction of N, let $N^a f$ be the chain map consisting of the arrows $\eta_n : N^a A_n \longrightarrow N^a B_n$ for each $n \geq -1$. η being a natural transformation ensures that this is a chain map, the same as in 2.5, with the extra square

$$\begin{array}{ccc} [-1] & & A_{-1} \longrightarrow B_{-1} \\ \downarrow & & {\scriptstyle \epsilon^A}\uparrow \qquad {\scriptstyle \epsilon^B}\uparrow \\ [0] & & A_0 \longrightarrow B_0 \end{array}$$

It is easy to check that N^a is a functor

$$N^a : \mathbf{sAb}^a \longrightarrow \mathbf{Ch}_{\geq -1}.$$

Based on the similarity of N^a compared to N, one can ask whether there is also an equivalence between $\mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathbf{Ab})$ and $\mathbf{Ch}_{\geq -1}(\mathbf{Ab})$.

Theorem 2.7 (Augmented Dold-Kan correspondence)**.**

$$N^a : \mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathbf{Ab}) \longrightarrow \boldsymbol{Ch}_{\geq -1}(\mathbf{Ab})$$

is an equivalence of categories.

In the next section we will demonstrate a proof of this theorem.

2.4 Proof of the augmented Dold-Kan correspondence

The proof will be conducted as follows. First we will prove the following three lemmas[3]. In their proof we will construct explicit categorical equivalences, P_1, P_2 and P_3. These equivalences will compose to give the functor N^a. This will complete the proof of Theorem 2.7. The second lemma is a consequence of a general result in comma categories, which we prove in Appendix A.3 (Lemma A.1). It is there that the functor P_2 is defined.

Lemma 2.8 (Identification 1)**.**

$$\mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathbf{Ab}) \simeq (\mathrm{Fun}(\mathbb{\Delta}, \mathbf{Ab}) \downarrow \mathbf{Ab}).$$

Lemma 2.9 (Dold-Kan on comma categories)**.**

$$(\mathrm{Fun}(\mathbb{\Delta}, \mathbf{Ab}) \downarrow \mathbf{Ab}) \simeq (\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}).$$

Lemma 2.10 (Identification 2)**.**

$$(\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}) \simeq \mathbf{Ch}_{\geq -1}(\mathbf{Ab}).$$

There is some abusive notation appearing in the statement of the lemmas, which will be explained at the start of each lemmas proof.

[3]Suggested in a different form by Lior Yanovski

2.4.1 Proof of Lemma 2.8

Consider the identity functor on $\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab})$. We also define the functor

$$\mathrm{const} : \mathbf{Ab} \longrightarrow \mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab})$$

which associates to an abelian group $A_{-1} \in \mathbf{Ab}$ the constant simplicial abelian group

$$A_\bullet : \quad [n] \mapsto A_{-1} \quad \text{and} \quad (f : [n] \longrightarrow [m]) \mapsto \mathrm{id} : A_{-1} \longrightarrow A_{-1}.$$

An arrow $f : A_{-1} \longrightarrow B_{-1}$ is sent to that natural transformation $\mathrm{const}(A_{-1}) \longrightarrow \mathrm{const}(B_{-1})$ consisting of the single component arrow f. We can therefore form the comma category

$$(\mathrm{id}_{\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab})} \downarrow \mathrm{const})$$

which we write

$$(\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab}) \downarrow \mathbf{Ab}).$$

In this section we will frequently change from indices $n \geq -1$ and $n \geq 0$ and choose to abusively write $A_\bullet = (A_n)_{n \geq 0}$ or $A_\bullet = (A_n)_{n \geq -1}$ where appropriate. Similarly a natural transformation $\eta : A_\bullet \longrightarrow B_\bullet$ contains component morphisms η_n and we may write $\eta = (\eta_n)_{n \geq -1}$ or $(\eta_n)_{n \geq 0}$ depending on the situation.

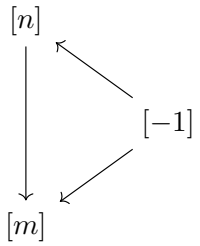
What does $(\mathrm{Fun}(\mathbb{\Delta}^{\mathrm{op}}, \mathbf{Ab}) \downarrow \mathbf{Ab})$ look like? Objects are triples $(A_\bullet, A_{-1}, A_\bullet \longrightarrow \mathrm{const}(A_{-1}))$. The arrow

$$A_\bullet \longrightarrow \mathrm{const}(A_{-1})$$

is a natural transformation, with components the horizontal arrows making

$$\begin{array}{ccc} [n] & A_n & \longrightarrow \mathrm{const}(A_{-1})[n] = A_{-1} \\ {\scriptstyle f}\downarrow & {\scriptstyle A_\bullet f}\uparrow & \uparrow{\scriptstyle \mathrm{id}} \\ [m] & A_m & \longrightarrow \mathrm{const}(A_{-1})[m] = A_{-1}. \end{array} \tag{2.6}$$

commute. Recall that in $\mathbb{\Delta}_\eta$, $[-1]$ is initial, hence for any $n, m \geq 0$,



commutes. Therefore if $A_\bullet = (A_n)_{n\geq -1} \in \mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathbf{Ab})$, we can always produce 2.6 and hence the object $((A_n)_{n\geq 0}, A_{-1}, (A_n)_{n\geq 0} \longrightarrow \mathrm{const}(A_{-1}))$ in the comma category. We are therefore lead to define the functor

$$\begin{gathered} P_1 : \mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathbf{Ab}) \longrightarrow (\mathrm{Fun}({\Delta}^{\mathrm{op}}, \mathbf{Ab}) \downarrow \mathbf{Ab}) \\ (A_n)_{n\geq -1} \mapsto ((A_n)_{n\geq 0}, A_{-1}, (A_n)_{n\geq 0} \longrightarrow \mathrm{const}(A_{-1})) \end{gathered}$$

sending a natural transformation $\eta = (\eta_n)_{n\geq -1} : (A_n)_{n\geq -1} \longrightarrow (B_n)_{n\geq -1}$ to the arrow

$$P_1((A_n)_{n\geq -1}) \longrightarrow P_1((B_n)_{n\geq -1})$$

in the comma category, consisting of the arrows

$$\begin{gathered} (\eta_n)_{n\geq 0} : (A_n)_{n\geq 0} \longrightarrow (B_n)_{n\geq 0} \\ \eta_{-1} : A_{-1} \longrightarrow B_{-1}. \end{gathered}$$

To check this is an arrow in the comma category, we need to check commutativity of

$$\begin{array}{ccc} (A_n)_{n\geq 0} & \longrightarrow & \mathrm{const}(A_{-1}) \\ {\scriptstyle (\eta_n)_{n\geq 0}}\downarrow & & \downarrow{\scriptstyle \mathrm{const}((\eta_{-1}))} \\ (B_n)_{n\geq 0} & \longrightarrow & \mathrm{const}(B_{-1}), \end{array} \tag{2.7}$$

where we are composing natural transformations here. One can show that this amounts to checking the commutativity of the square

$$\begin{array}{ccc} A_n & \longrightarrow & A_{-1} \\ {\scriptstyle \eta_n}\downarrow & & \downarrow{\scriptstyle \eta_{-1}} \\ B_n & \longrightarrow & B_{-1} \end{array} \tag{2.8}$$

for all $n \geq 0$. Here the horizontal arrows correspond to $[-1] \longrightarrow [n]$ in Δ_η. By 2.6, 2.8 always commutes.

To establish the desired equivalence we need to verify that P_1 is (1) functorial, (2) fully faithful and (3) essentially surjective[4].

1. Let $A_\bullet \xrightarrow{\phi} B_\bullet \xrightarrow{\eta} C_\bullet$, where $\phi = (\phi_n)_{n\geq -1}$ and $\eta = (\eta_n)_{n\geq -1}$. Then both $P_1(\eta\phi)$ and $P_1(\eta) \circ P_1(\phi)$ are the arrow $P_1(A_\bullet) \longrightarrow P_1(C_\bullet)$ consisting of the arrows

$$\begin{gathered} (\eta_n\phi_n)_{n\geq 0} : (A_n)_{n\geq 0} \longrightarrow (C_n)_{n\geq 0} \\ \eta_{-1}\phi_{-1} : A_{-1} \longrightarrow C_{-1}. \end{gathered}$$

[4]We are assuming the axiom of choice.

2. Fix $A_\bullet$ and $B_\bullet$ in $\mathrm{Fun}(\Delta_\eta{}^{\mathrm{op}}, \mathbf{Ab})$. We need to demonstrate a bijection

$$\mathrm{Hom}(A_\bullet, B_\bullet) \longrightarrow \mathrm{Hom}(P_1 A_\bullet, P_1 B_\bullet).$$

But this follows from the definition of P_1. Any natural transformation $\eta = (\eta_n)_{n\geq -1} : A_\bullet \longrightarrow B_\bullet$ is uniquely broken up into $(\eta_n)_{n\geq 0}$ and η_{-1} to form an arrow $P_1 A_\bullet \longrightarrow P_1 B_\bullet$. Conversely any $(\eta_n)_{n\geq 0}$ and η_{-1} can be glued together to form a natural transformation $(\eta_n)_{n\geq -1}$. To see this second claim we only need to check naturality on an arrow $[-1] \longrightarrow [n]$ in Δ_η, $n \geq 0$. That is, commutativity of

$$\begin{array}{ccc} A_n & \longrightarrow & A_{-1} \\ {\scriptstyle \eta_n}\downarrow & & \downarrow{\scriptstyle \eta_{-1}} \\ B_n & \longrightarrow & B_{-1} \end{array}$$

This is implied by commutativity of 2.7, which follows since P_1 is a functor. We have demonstrated the desired bijection.

3. Fix $((A_n)_{n\geq 0}, A, (A_n)_{n\geq 0} \longrightarrow \mathrm{const}(A))$ in $(\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab}) \downarrow \mathbf{Ab})$. It is enough to show that $(A_n)_{n\geq -1}$ (where $A_{-1} := A$) is a functor in $\mathrm{Fun}(\Delta_\eta{}^{\mathrm{op}}, \mathbf{Ab})$. Define the functor

$$\begin{aligned} A_\bullet = (A_n)_{n\geq -1} : \Delta_\eta{}^{\mathrm{op}} &\longrightarrow \mathbf{Ab} \\ [n] &\mapsto A_n \ (n \geq 0) \\ [-1] &\mapsto A =: A_{-1} \end{aligned}$$

and an arrow $[n] \longrightarrow [m]$ is sent to $A_m \longrightarrow A_n$ for $m, n \geq 0$. For $n \geq 0$, let the arrow $[-1] \longrightarrow [n]$ be sent to the corresponding component in the natural transformation $(A_n)_{n\geq 0} \longrightarrow \mathrm{const}(A)$. We need to check functorality.

Let $[l] \longrightarrow [n] \longrightarrow [m]$. If $l, m, n \geq 0$ then $A_\bullet$ preserves this composition since we already had a functor $(A_n)_{n\geq 0} \in \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab})$. We need to consider what happens to $[-1] \longrightarrow [n] \longrightarrow [m]$ when $n, m \geq 0$. (Note that $[-1]$ is initial in Δ_η, so can only appear at the start of a string of arrows.) We need to check that

$$A_m \longrightarrow A_n \longrightarrow A_{-1} \quad \text{equals} \quad A_m \longrightarrow A_{-1}.$$

This was shown in 2.6.

The proof of Lemma 2.8 is complete.

2.4.2 Proof of Lemmas 2.9 and 2.10

We need to demonstrate the equivalences

$$\begin{aligned}(\mathrm{Fun}(\mathbb{\Delta}, \mathbf{Ab}) \downarrow \mathbf{Ab}) &\simeq (\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}) \quad \text{(Lem. 2.9)} \\ &\simeq \mathbf{Ch}_{\geq -1}(\mathbf{Ab}) \quad \text{(Lem. 2.10).}\end{aligned}$$

First note that by $(\mathbf{Ch}_{\geq -1}(\mathbf{Ab}) \downarrow \mathbf{Ab})$ we mean $(\mathrm{id}_{\mathbf{Ch}_{\geq -1}(\mathbf{Ab})} \downarrow (-)[0])$ where

$$(-)[0] : \mathbf{Ab} \longrightarrow \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$

is the functor sending

$$\mathbf{Ab} \ni A \mapsto A[0] = (A \longleftarrow 0 \longleftarrow 0 \longleftarrow \dots) \in \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$

and a group homomorphism $f : A \longrightarrow B$ to the chain map

$$\begin{array}{ccccccc} A & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots \\ {\scriptstyle f}\downarrow & & \downarrow & & \downarrow & & \\ B & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots \end{array}$$

We denote this chain map by $\tilde{f}$. It is easy to check functorality.

By Lemma A.1, the classical Dold-Kan correspondence gives us the first equivalence. To demonstrate the second, define a functor

$$\begin{aligned} Q : \mathbf{Ch}_{\geq -1}(\mathbf{Ab}) &\longrightarrow (\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}) \\ A_\bullet = (A_n, d_{n+1})_{n \geq -1} &\mapsto ((A_n, d_{n+1})_{n \geq 0}, A_{-1}, \tilde{d}_0) \end{aligned}$$

which sends an arrow $f_\bullet : A_\bullet \longrightarrow B_\bullet$ to the arrow

$$((A_n, d^A_{n+1})_{n \geq 0}, A_{-1}, \tilde{d}^A_0) \longrightarrow ((B_n, d^B_{n+1})_{n \geq 0}, B_{-1}, \tilde{d}^B_0)$$

consisting of the maps

$$(f_n)_{n \geq 0} : (A_n, d^A_{n+1})_{n \geq 0} \longrightarrow (B_n, d^B_{n+1})_{n \geq 0} \tag{2.9}$$

$$f_{-1} : A_{-1} \longrightarrow B_{-1}. \tag{2.10}$$

To verify that this is an arrow in $(\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab})$, we need to check commutativity of the square

$$\begin{array}{ccc} (A_n, d^A_{n+1})_{n \geq 0} & \xrightarrow{\tilde{d}^A_0} & A_{-1}[0] \\ {\scriptstyle (f_n)_{n \geq 0}}\downarrow & & \downarrow{\scriptstyle \tilde{f}_{-1}} \\ (B_n, d^B_{n+1})_{n \geq 0} & \xrightarrow[\tilde{d}^B_0]{} & B_{-1}[0] \end{array} \tag{2.11}$$

Chasing around the top we get

$$\begin{array}{ccccccc} A_0 & \longleftarrow & A_1 & \longleftarrow & A_2 & \longleftarrow & \dots \\ {\scriptstyle d_0^A}\downarrow & & \downarrow & & \downarrow & & \\ A_{-1} & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots \\ {\scriptstyle f_{-1}}\downarrow & & \downarrow & & \downarrow & & \\ B_{-1} & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots \end{array}$$

The composed component chain maps are zero in non-negative degrees. We get a similar result chasing around the bottom of 2.11. It remains to check equality of the chain maps in degree -1. That is, equality of

$$f_{-1}d_0^A,\ d_0^B f_0 : A_0 \longrightarrow B_{-1}.$$

But this follows since we originally had the chain map $f_\bullet : A_\bullet \longrightarrow B_\bullet$.

To achieve the second equivalence (Lemma 2.10) we need to show Q is (1) functorial, (2) fully faithful and (3) essentially surjective.

1. For functorality, consider arrows

   $$A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet$$

   where $A_\bullet = (A_n, d_{n+1}^A)_{n\geq -1}$, and similar for $B_\bullet$ and $C_\bullet$. $Q(g \circ f)$ is then the arrow

   $$((A_n, d_{n+1}^A)_{n\geq 0}, A_{-1}, \tilde{d}_0^A) \longrightarrow ((C_n, d_{n+1}^C)_{n\geq 0}, C_{-1}, \tilde{d}_0^C)$$

   consisting of the arrows

   $$(g_n \circ f_n)_{n\geq 0} : (A_n, d_{n+1}^A)_{n\geq 0} \longrightarrow (C_n, d_{n+1}^C)_{n\geq 0}$$
   $$g_{-1} \circ f_{-1} : A_{-1} \longrightarrow C_{-1},$$

   which is clearly equal to $Q(g) \circ Q(f)$ (see Definition A.4).

2. For fully faithful, fix $A_\bullet$ and $B_\bullet$ in $\mathbf{Ch}_{\geq -1}(\mathbf{Ab})$. We need to demonstrate a bijection

   $$\mathrm{Hom}(A_\bullet, B_\bullet) \longrightarrow \mathrm{Hom}(QA_\bullet, QB_\bullet).$$

   But this follows from examining 2.9, 2.10 and 2.11. In particular, 2.11 ensures $(f_n)_{n\geq 0}$ and f_{-1} form a chain map $A_\bullet \longrightarrow B_\bullet$.

3. Consider an object $((A_n, d_{n+1})_{n\geq 0}, A, (f_n)_{n\geq 0})$ in $(\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab})$. We have

$$\begin{array}{ccccccccc} A_0 & \xleftarrow{d_0} & A_1 & \xleftarrow{d_1} & A_2 & \xleftarrow{d_2} & \dots \\ {\scriptstyle f_0}\downarrow & & \downarrow & & \downarrow & & \\ A & \longleftarrow & 0 & \longleftarrow & 0 & \longleftarrow & \dots \end{array}$$

which shows the composition $f_0 \circ d_0$ is zero. Essential surjectivity follows.

The proof of lemmas 2.9 and 2.10 are complete.

Remark 1. One can show that the inverse to Q in Lemma 2.10 is the functor

$$\begin{gathered} P_3 : (\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}) \longrightarrow \mathbf{Ch}_{\geq -1}(\mathbf{Ab}) \\ ((A_n)_{n\geq 0}, A, (f_n)_{n\geq 0} : (A_n)_{n\geq 0} \longrightarrow A[0]) \mapsto (A_n)_{n\geq -1}, \end{gathered}$$

which sends an arrow

$$((A_n)_{n\geq 0}, A_{-1}, f : (A_n)_{n\geq 0} \longrightarrow A[0]) \longrightarrow ((B_n)_{n\geq 0}, B_{-1}, g : (B_n)_{n\geq 0} \longrightarrow B[0])$$

consisting of the arrows

$$\begin{gathered} (\xi_n)_{n\geq 0} : (A_n)_{n\geq 0} \longrightarrow (B_n)_{n\geq 0} \\ \xi_{-1} : A_{-1} \longrightarrow B_{-1} \end{gathered}$$

to the chain map $(\xi_n)_{n\geq -1} : (A_n)_{n\geq -1} \longrightarrow (B_n)_{n\geq -1}$.

We will use P_3 below.

2.4.3 Proof that N^a is an equivalence

We will show the above equivalences compose to give N^a. That is,

$$\begin{aligned} \mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathbf{Ab}) &\xrightarrow{\simeq} (\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab}) \downarrow \mathbf{Ab}) && \text{(Lem. 2.8)} \\ &\xrightarrow{\simeq} (\mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \downarrow \mathbf{Ab}) && \text{(Lem. A.1 and Thm. 2.6)} \\ &\xrightarrow{\simeq} \mathbf{Ch}_{\geq -1}(\mathbf{Ab}) && \text{(Rmk. 1)} \end{aligned}$$

compose to give N^a. We need to demonstrate this equality of functors on objects and arrows. First for arrows, let $A_\bullet = (A_n)_{n\geq -1} \in \mathrm{Fun}({\Delta_\eta}^{\mathrm{op}}, \mathbf{Ab})$. The first functor takes $A_\bullet$ to

$$((A_n)_{n\geq 0}, A_{-1}, (A_n)_{n\geq 0} \longrightarrow \mathrm{const}(A_{-1})).$$

This then becomes

$$(N(A_n)_{n\geq 0}, A_{-1}, N(A_n)_{n\geq 0} \longrightarrow N\,\mathrm{const}(A_{-1})) \tag{2.12}$$

N being the normalised chain complex functor from Theorem 2.6. $\mathrm{const}(A_{-1})$ is the simplicial object

$$[n] \mapsto A_{-1}, \quad ([n] \longrightarrow [m]) \mapsto (\mathrm{id} : A_{-1} \longrightarrow A_{-1}).$$

The normalised complex $\mathrm{const}(A_{-1})$ is

$$(A_{-1} \longleftarrow 0 \longleftarrow 0 \longleftarrow \dots) = A_{-1}[0]$$

of all zero maps. 2.12 is finally sent to

$$A_{-1} \xleftarrow{\epsilon} A_0 \xleftarrow{(-1)d_1} NA_1 \longleftarrow \dots \longleftarrow NA_{n-1} \xleftarrow{(-1)^n d_n} NA_n \longleftarrow \dots$$

Checking what happens to arrows is simpler. We need to check that a natural transformation $\eta = (\eta_n)_{n\geq -1} : A_\bullet \longrightarrow B_\bullet$ in $\mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathbf{Ab})$ is sent to the chain map $(A_n)_{n\geq -1} \longrightarrow (B_n)_{n\geq -1}$ consisting of arrows $\eta_n : N^a A_n \longrightarrow N^a B_n$.

$$\begin{aligned}(\eta_n)_{n\geq -1} &\mapsto ((\eta_n)_{n\geq 0}, \eta_{-1}) \\ &\mapsto (N(\eta_n)_{n\geq 0}, \eta_{-1}) = ((\eta_n)_{n\geq 0}, \eta_{-1}) \quad \text{(now as a chain map on } N^a(A_n)_{n\geq 0}) \\ &\mapsto (\eta_n)_{n\geq -1}\end{aligned}$$

as desired. Theorem 2.7 is proven.

2.5 Extensions of the augmented Dold-Kan correspondence

We will now discuss extensions of the augmented Dold-Kan correspondence.

It is worth asking how the equivalence of categories

$$\mathrm{Fun}(\mathbb{\Delta}_\eta{}^{\mathrm{op}}, \mathbf{Ab}) \simeq \mathbf{Ch}_{\geq -1}(\mathbf{Ab}) \tag{2.13}$$

can be extended from $\mathbf{Ab}$ to more general categories.

One can ask about promoting from $\mathbf{Ab}$ to abelian categories. Abelian categories are as general as one goes to define chain complexes, talk about kernels and, therefore, ask about analogues of Dold-Kan. Indeed, the Dold-Kan correspondence still holds in abelian categories [DP61]. One needs to be careful when working with abelian categories, since they are not always concrete. (For some intuition on what this means, see [Fre19].) In particular, we need to verify that the arguments we make are not at the level of *elements* in objects in our category. Instead arguments should be made at the level of arrows. This is the case here and one can expect the augmented correspondence to hold for abelian categories.[5]

A result for the classical Dold-Kan correspondence is that it extends to a Quillen equivalence [Qui67]. In particular, N induces an equivalence of the homotopy categories

[5]It was pointed out to me that there may be a way around this, since our diagrams are *small*.

in Dold-Kan. This has implications in homotopy theory, see [GJ12]. We conjecture that the augmented correspondence also extends to a Quillen equivalence.

In [GJ12] it was noted that an adjoint to N, Γ (a nerve functor), can be used to compute Eilenberg-MacLane spaces. In particular, if $A[n]$ is the chain complex with $A \in \mathbf{Ab}$ concentrated in degree n, then $\Gamma A[n] = K(A, n)$. Moreover, if $B_\bullet$ is a simplicial abelian group, then

Theorem 2.11 ([GJ12])**.**

$$[B_\bullet, K(A, n)] \cong H^n(B_\bullet, A) \quad (n \geq 0)$$

where $[-,-] = [-,-]_{\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab})}$ *represents the simplicial homotopy classes and* $B_\bullet$*, on the right, is identified with its Moore complex.*

One can ask about the existence of similar isomorphisms in the augmented world.

A Category theory

A.1 Monoidal categories

The reader is encouraged to reference the article [Bae04].

Definition A.1. A *monoidal* category M is a category, together with the following.

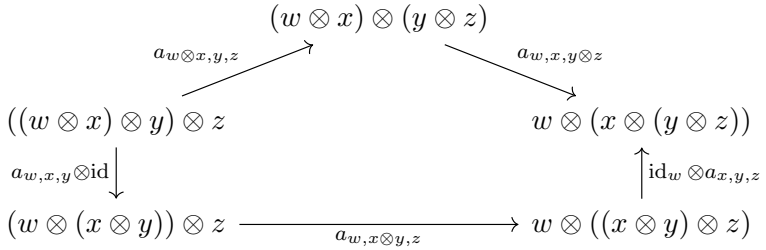
1. a *bifunctor* $\otimes : M \times M \to M$ (write $x \otimes y$ and $f \otimes g$ for the image of two objects and morphisms under $\otimes$) called the *tensor product*.
2. an identity object $1 \in M$.
3. Natural isomorphisms

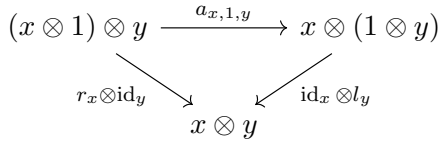
$$a_{x,y,z} : (x \otimes y) \otimes z \xrightarrow{\sim} x \otimes (y \otimes z)$$

and

$$l_x : 1 \otimes x \xrightarrow{\sim} x, \quad r_x : x \otimes 1 \xrightarrow{\sim} x.$$

for objects $x, y, z \in M$. These natural isomorphisms are required to satisfy the following *coherence* conditions (called the *pentagon* and *triangle* equations).





We will write $(M, \otimes, 1)$ to represent the above data. M is called strict strict if the natural isomorphisms are identities. In this case, $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ and $1 \otimes x = x = x \otimes 1$.

Remark 2. Strict monoidal categories are rare. Vector spaces with the usual tensor product are, for example, are not strictly monoidal.

A.2 Composition of functors

Let $F, G, H : \mathcal{C} \to \mathcal{D}$ be functors and $\mu : F \implies G$ and $\eta : G \implies H$ natural transformations. Consider any $f : x \to y$ in $\mathcal{C}$. The natural transformations give us commutative squares

$$\begin{array}{ccccc} x & & F(x) & \longrightarrow & G(x) \longrightarrow H(x) \\ f\downarrow & & \downarrow F(f) & & \downarrow G(f) \qquad \downarrow H(f) \\ y & & F(y) & \longrightarrow & G(y) \longrightarrow H(y) \end{array}$$

Composing component transformations to produce the above dotted arrows gives the commutative square

$$\begin{array}{ccc} x & F(x) & \longrightarrow H(x) \\ f\downarrow & \downarrow F(f) & \qquad \downarrow H(f) \\ y & F(y) & \longrightarrow H(y) \end{array}$$

This gives a natural transformation $F \implies H$.

Definition A.2. Let $F, G, H : \mathcal{C} \to \mathcal{D}$ be functors and $\mu : F \implies G$ and $\eta : G \implies H$ natural transformations. *Vertical composition* of μ, η is the natural transformation

$$\eta \circ \mu : F \implies H$$

(also written $\eta\mu$) given by composing component arrows, as above.

Remark 3. The term *vertical* composition refers to composition of functors between the same categories, where the data can be arranged as the diagram



Remark 4. Vertical composition of natural transformations is *strictly* associative, since the composition of component arrows is strictly associative.

A.3 Comma categories

One should read either [Rie16, Chapter 1] or [Mac13, Chapter II.6] for more on comma categories.

Definition A.3 ([Rie16])**.** Let

$$F : \mathcal{D} \longrightarrow \mathcal{C} \longleftarrow \mathcal{E} : G$$

be functors between categories. The *comma category* $(F \downarrow G)$ is the category with

1. Objects as triples (d, e, f) where $d \in \mathcal{D}, e \in \mathcal{E}$ and $f : Fd \longrightarrow Ge$ is an arrow in $\mathcal{C}$.
2. Arrows $(d, e, f) \longrightarrow (d', e', f')$ are a pair of arrows

$$h : d \longrightarrow d', \quad k : e \longrightarrow e'$$

such that

$$\begin{array}{ccc} Fd & \xrightarrow{f} & Ge \\ {\scriptstyle Fh}\downarrow & & \downarrow{\scriptstyle Gk} \\ Fd' & \xrightarrow[f']{} & Ge' \end{array}$$

commutes.

Definition A.4. Let $(F \downarrow G)$ (where $F : \mathcal{D} \longrightarrow \mathcal{C} \longleftarrow \mathcal{E} : G$) be a comma category. The composition of two arrows

$$(d, e, f) \xrightarrow[\substack{k:e\longrightarrow e'}]{h:d\longrightarrow d'} (d', e', f') \xrightarrow[\substack{k':e'\longrightarrow e''}]{h':d'\longrightarrow d''} (d'', e'', f'')$$

is defined by

$$(d, e, f) \xrightarrow[\substack{k'\circ k:e\longrightarrow e''}]{h'\circ h:d\longrightarrow d''} (d'', e'', f'').$$

In simpler words, composition is given by composing the two morphisms associated to each arrow.

Remark 5. The composition of arrows is well defined since the inner squares in

$$\begin{array}{ccccc} & Fd & \xrightarrow{f} & Ge & \\ & {\scriptstyle Fh}\downarrow & & \downarrow{\scriptstyle Gk} & \\ F(h'\circ h) & Fd' & \xrightarrow[f']{} & Ge' & G(k'\circ k) \\ & {\scriptstyle Fh'}\downarrow & & \downarrow{\scriptstyle Gk'} & \\ & Fd'' & \xrightarrow[f'']{} & Ge'' & \end{array}$$

commute and F and G are functors.

In the proof of the augmented Dold-Kan correspondence (Theorem 2.7), we need the following lemma for general comma categories. Let $\mathcal{C}, \mathcal{D}$ and $\mathcal{E}$ be categories and $G : \mathcal{D} \longrightarrow \mathcal{C}$ a functor. Consider the comma category $\mathrm{id}_{\mathcal{C}} \downarrow G$ which we abusively denote $\mathcal{C} \downarrow G$. Recall from Definition A.3 that arrows $(x, y, f : x \longrightarrow y) \longrightarrow (x', y', f' : x' \longrightarrow y')$ in $(\mathcal{C} \downarrow G)$ are two arrows $h : x \longrightarrow x', k : y \longrightarrow y'$ s.t. the diagram

$$\begin{array}{ccc} x & y & x \xrightarrow{f} Gy \\ {\scriptstyle h}\downarrow & {\scriptstyle k}\downarrow & {\scriptstyle h}\downarrow \quad\quad \downarrow{\scriptstyle Gk} \\ x' & y' & x' \xrightarrow[f']{} Gy' \end{array}$$

commutes.

Lemma A.1. *Let $G : \mathcal{D} \longrightarrow \mathcal{C}$ be a functor and $\eta : \mathcal{C} \xrightarrow{\simeq} \mathcal{E}$ a functor that is an equivalence of categories. Then,*

$$(\mathcal{C} \downarrow G) \simeq (\mathcal{E} \downarrow \eta \circ G).$$

Proof. Let

$$P_2 : (\mathcal{C} \downarrow G) \longrightarrow (\mathcal{E} \downarrow \eta \circ G)$$
$$(x, y, f : x \longrightarrow Gy) \mapsto (\eta x, y, \eta f : \eta x \longrightarrow \eta G y)$$

where an arrow $(x, y, f : x \longrightarrow y) \longrightarrow (x', y', f' : x' \longrightarrow y')$, i.e., two arrows $h : x \longrightarrow x', k : y \longrightarrow y'$, is sent to

$$\eta h : \eta x \longrightarrow \eta x', \quad k : y \longrightarrow y'.$$

Indeed this is an arrow in $(\mathcal{E} \downarrow \eta G)$ since η preserves diagrams. P_2 is functorial since if

$$(d, e, f) \xrightarrow[\substack{k:e\longrightarrow e'}]{\substack{h:d\longrightarrow d'}} (d', e', f') \xrightarrow[\substack{k':e'\longrightarrow e''}]{\substack{h':d'\longrightarrow d''}} (d'', e'', f'') \qquad \overset{\overset{h'\circ h:d\longrightarrow d''}{k'\circ k:e\longrightarrow e''}}{\frown}$$

is a composition of arrows, then under P_2 we get

$$(\eta d, e, \eta f) \xrightarrow[\substack{k:e\longrightarrow e'}]{\substack{\eta(h):\eta d\longrightarrow \eta d'}} (\eta d', e', \eta f') \xrightarrow[\substack{k':e'\longrightarrow e''}]{\substack{\eta(h'):\eta d'\longrightarrow \eta d''}} (\eta d'', e'', \eta f'') \qquad \overset{\overset{\eta(h'\circ h):\eta d\longrightarrow \eta d''}{k'\circ k:e\longrightarrow e''}}{\frown}$$

which clearly commutes since η is functorial. It remains to show P_2 is (1) fully faithful and (2) essentially surjective.

1. P_2 is fully faithful. This follows since η is fully faithful. Namely, η gives a bijection $h \mapsto \eta h$ (of hom sets in $\mathcal{C}$ and $\mathcal{E}$), which then extends to give that $(h, k) \mapsto (\eta h, k)$ is a bijection between the hom sets of $(\mathcal{C} \downarrow G)$ and $(\mathcal{E} \downarrow \eta G)$.

2. P_2 is essentially surjective. This follows since η is both essentially surjective and fully faithful. Essentially surjective gives us surjectivity, up to isomorphism, of $x \mapsto \eta x$. Fully faithful gives us a bijection $f \mapsto \eta f$ of hom sets in $\mathcal{C}$ and $\mathcal{E}$. Combining these two gives us surjectivity, up to isomorphism, of $(x, y, f) \mapsto (\eta x, y, \eta f)$ (applying the isomorhpism on the ηx entry).

We conclude now that P_2 is an equivalence of categories. $\square$

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